

BRIAN KURIA MAINA

CTO & Full-Stack Product Engineer

AI Platforms · Flutter Mobile · Django / Next.js Backends

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Nairobi, Kenya

[Portfolio](#) · [GitHub](#) · [CBC AI Platform](#) · [CBC AI on Play Store](#) · [Tuko Kadi on Play Store](#)

PROFESSIONAL SUMMARY

CTO and full-stack product engineer who ships production systems end to end — mobile, backend, data and deployment. Co-architect of **CBC AI**, an AI infrastructure platform for Kenyan schools now live in production with **1,000+ Play Store downloads**, where I own the mobile application layer. Independently designed, built and deployed three further production systems: a real-money **M-Pesa B2C payroll and farm ERP (151k+ lines, live)**, an offline-first **peer-to-peer multiplayer card game** with an embedded WebSocket host and AI opponents (55k+ lines, shipped), and a **multi-timeframe financial analytics dashboard** on serverless Postgres. Strong bias toward context-rich AI systems, zero-infrastructure architectures and products that work in low-connectivity environments.

CORE SKILLS

Languages	Dart, Python, JavaScript (ES6+), TypeScript, SQL
Mobile	Flutter 3.x, cross-platform Android & iOS, Play Console release management, AdMob, In-App Purchase / Play Billing
Backend	Django 5, Django REST Framework, FastAPI, Node.js, REST API design, webhook & callback architecture
Frontend	Next.js 15 (App Router), React 19, Tailwind CSS, Framer Motion, Radix UI, Recharts, Chart.js
Data	PostgreSQL (Neon serverless), MySQL, SQLite, Drizzle ORM, Django ORM, schema design & aggregation queries
AI & Intelligence	AI-assisted product workflows, RAG concepts, knowledge-graph modelling, heuristic decision engines, behavioural bot AI
Integrations	Safaricom Daraja M-Pesa B2C, Brevo transactional email, Clerk Auth, bulk SMS, QR pairing, WebSockets

DevOps	cPanel / Phusion Passenger, Railway, Vercel, WhiteNoise, Gunicorn, Git & GitHub, Postman, Flutter DevTools
Practices	System architecture, state-machine design, RBAC, automated simulation testing, technical documentation

EXPERIENCE

Chief Technology Officer — Bombay Softwares · Nairobi, Kenya

2023 – Present

- Set technical direction for an EdTech startup building **AI infrastructure for schools**, not point-solution school software — every workflow is designed to generate educational context that AI can reason over.
- Co-developed the **CBC AI Platform** (cbcai.co.ke) alongside a partner team, owning the **mobile application layer** end to end — architecture, UI/UX, state management, API integration, release engineering and store operations.
- Shipped and maintain the **CBC AI Android & iOS app** to production, now at **1,000+ downloads** and in active daily use by Kenyan schools.
- Architected the app against a four-layer platform model — data collection (school workflows, hardware/IoT), educational intelligence (learner & teacher profiles, school knowledge graph), AI (Teacher AI, Learner AI, Parent AI, School AI) and measurable impact.
- Translated the Competency-Based Curriculum into product: CBC-aligned assessment reporting, competency tracking, learner progress records and teacher administrative automation (lesson plans, schemes of work, records of work, report cards).
- Drove REST API contracts and performance optimisation with backend engineers; own the full release pipeline from development through QA to Google Play Store deployment.
- Apply a single build filter across the roadmap: *does this generate context that makes teachers or learners more productive?*

ICT Attachment — Office of the Attorney General & Department of Justice · Nairobi, Kenya

May 2026 – Aug 2026

- Managed the **ICT helpdesk** for the State Law Office and Department of Justice, resolving user hardware, software and network support requests.
- Configured and troubleshoot **LAN/Internet devices** to keep departmental network connectivity reliable.
- Administered **Active Directory** user accounts on the MS-Windows Server Domain.
- Installed and configured **operating systems, antivirus and user programs** across staff workstations.
- Carried out **preventive maintenance and minor repairs** of ICT equipment.
- Configured, managed and maintained office **printers**.
- **Backed up and recovered** departmental data.

- Supported **digitisation and data entry** initiatives for departmental records.

SELECTED PROJECTS

CBC AI — AI Infrastructure Platform for Schools

 *Partnered product · Role: Co-developer & CTO, mobile lead · Live in production · 1,000+ downloads*

Stack: Flutter · Dart · Django REST Framework · PostgreSQL · REST APIs · Bulk SMS · RFID / biometric hardware integrations

[Platform](#) · [Play Store](#)

A production AI platform that turns everyday school operations into structured context, then uses that context to cut teacher workload and personalise learning.

- Own the **mobile app layer**: cross-platform Flutter delivery for Android and iOS, from architecture and design system through release and post-launch iteration.
- Surfaced the platform's intelligence layer on mobile — learner profiles, competency growth, CBC-aligned assessment reporting and teacher dashboards for class and student management.
- Shipped school operations modules used daily: attendance, fee management and financial tracking, timetables, records of work, and teacher–parent SMS communication.
- Connected the app to a data collection layer spanning school workflows and hardware/IoT signals (RFID canteen payments, biometric attendance, library RFID) that feed the platform's school knowledge graph.
- Positioned mobile as the primary delivery surface for Teacher AI (lesson plans, schemes of work, assessments, marking, reports, interventions) and Learner AI (strengths discovery, competency growth, study plans, career exploration).

Tuko Kadi — Offline-First P2P Multiplayer Card Game

 Solo project · Role: Sole engineer & game architect · Shipped v1.0.0+12 · May 2026 – Aug 2026

Stack: Flutter 3.x · Dart 3.x · embedded dart:io HttpServer + WebSockets · CustomPainter · AdMob · In-App Purchase · shared_preferences · flutter_test

[Play Store](#)

A console-quality mobile adaptation of the Kenyan card game Kadi, playable with zero internet and zero backend cost. 55,650+ lines of Dart across 86 files and 294+ commits, 100% solo-owned.

- Built a deterministic game engine and state machine (GameController, ValidationEngine, TurnManager, DeckManager) handling cascading card combinations, penalty stacking, Jack deflections and jumps, Ace suit calls and Super Ace demands.
- Engineered zero-infrastructure P2P multiplayer by embedding an HttpServer + WebSocket host directly on-device — players join by scanning a QR code carrying the host's local IP and room code, achieving **sub-5 ms local latency at zero cloud cost**.
- Made networking fault-tolerant with persistent clientId session recovery, automated reconnection, turn-timeout fail-safes and a clean lobby state-sync protocol that survives players backgrounding their phones.
- Solved asynchronous out-of-turn rule interruptions by isolating a pure ValidationEngine behind a negotiation state machine that freezes turns to dispatch interrupt prompts before resolving flow direction and penalties.
- Built four behavioural AI opponents (The Meta, Simba, Kobe, Fisi) and a headless simulator that runs 500+ full 4-player matches in seconds to verify zero rule violations, deck integrity and bot win-rate balance.
- Engineered an interactive spotlight onboarding system using GlobalKey render-box geometry with post-frame layout validation and fallback bounds, rendering correctly across all screen resolutions.
- Hand-built the visual layer: 10 curated dark themes, 14 animated app bars with glowing accent hairlines, custom-canvas avatar builder, card-shatter particle physics and celebration animations.
- Shipped a complete monetisation and store pipeline: AdMob banner/interstitial, IAP premium unlock and tipping tiers, a BillingGuard error-handling layer, and native splash screens — passing flutter analyze with zero issues.

Tea Farm Management System (TFMS) — Agricultural ERP with Real-Money Payouts

 Solo project · Role: Lead full-stack engineer & system architect · Live in production · Sept 2025 – Present

Stack: Python 3.13 · Django 5.2.6 · MySQL · Vanilla JS (ES6+) · Safaricom Daraja M-Pesa B2C · Brevo · openpyxl · cPanel / Phusion Passenger · WhiteNoise

[GitHub](#)

A full farm ERP replacing paper ledgers, phone-call coordination and risky cash payouts across tea estate operations. 151,800+ lines across 290+ files and 320+ commits, 100% solo-owned.

- Cut payroll calculation and wage disbursement from **2–3 business days to under 5 minutes** by automating harvest weight → piece-rate wage → mobile payout as one pipeline.
- Integrated Safaricom Daraja M-Pesa B2C for one-click wage, contractor and supply disbursements with **100% automated reconciliation** between Safaricom transaction IDs and internal ledger records, removing cash handling and manual entry.
- Solved multi-tenant webhook routing — Safaricom B2C callbacks carry no application state — by minting dynamic callback URLs with embedded farm IDs at dispatch time, then matching OriginatorConversationID in idempotent handlers that write immutable audit-ledger entries.
- Diagnosed and fixed a silent production data-loss bug where Django's `__date` lookups invoked MySQL `CONVERT_TZ()` against unpopulated shared-hosting timezone tables, returning NULL and rendering every dashboard empty; refactored date-boundary math into Python with `timezone.make_aware()` and exact UTC range filtering, making queries host-independent.
- Built 4-tier RBAC with distinct portals and permission boundaries for SuperAdmin, Farm Owner, Farm Manager and Field Worker.
- Delivered the operational core: harvest weighing logs (gross/tare/net kg with quality deductions), worker clock-in/out with auto-clockout sync, a conflict-detecting task scheduler preventing double-booked shifts across zones, and inventory tracking for equipment and agro-supplies.
- Added a role-based messaging hub with delivered/seen receipts, tokenised invitation onboarding over Brevo transactional email, and a reporting suite with financial analytics, productivity dashboards and dynamic Excel export.
- Deployed and maintain production on cPanel via Phusion Passenger with MySQL, WhiteNoise static delivery and cross-origin hardening.

Finance Tracker — Multi-Timeframe Financial Intelligence Dashboard

 Solo project · Role: Lead full-stack developer & UI/UX · Shipped MVP · Nov 2024

Stack: Next.js 15 (App Router, Turbopack) · React 19 · Neon serverless PostgreSQL · Drizzle ORM · Clerk Auth · Recharts · Tailwind CSS · Framer Motion · jsPDF / html2canvas · Vercel

[Live Site](#)

A localised (KES) personal finance dashboard built around flexible, overlapping time periods rather than fixed months.

- Designed a normalised 5-table serverless Postgres schema (periods, period_selected, budgets, incomes, expenses) with Drizzle ORM.
- Built a multi-timeframe state engine — a persistent TimeFrameProvider React Context backed by a period_selected table — letting users aggregate weekly, monthly, yearly or custom periods concurrently with no stale or hydration-mismatched data.
- Wrote dynamic Drizzle queries using inArray filtering and COALESCE aggregate joins to compute budget-vs-spend across arbitrary period selections at **sub-second response times**.
- Engineered an 11-scenario heuristic advisory engine that evaluates live income/budget/spend ratios to produce contextual recommendations and colour-coded risk alert states.
- Built an auto-cycling analytics dashboard (bar, line and dual pie charts for spend distribution and budget allocation) with custom tooltips and a 20-second carousel.
- Solved cross-device chart export with a reusable useChartExport hook that normalises container styles, captures at 3× resolution via html2canvas, computes A4-landscape aspect scaling and embeds into jsPDF — one-click, publication-ready reports in under a second.
- Secured routes with Clerk authentication and Next.js middleware; deployed on Vercel.

KEY ACHIEVEMENTS

- **CTO of an EdTech startup** while completing an undergraduate degree, co-owning an AI platform live in Kenyan schools.
- **Shipped two apps to the Google Play Store**, one at 1,000+ downloads and in active institutional use.
- **Built and deployed four production systems** across mobile, web, backend, hardware-adjacent and payments domains.
- **Handled real money in production** — M-Pesa B2C disbursements with fully automated reconciliation and immutable audit ledgers.
- **Authored 200,000+ lines of production code solo** across the Tuko Kadi and TFMS codebases alone.
- **Engineered a zero-backend real-time multiplayer architecture**, eliminating cloud costs while working entirely offline.

EDUCATION

BSc Software Development — KCA University, Nairobi, Kenya · *Expected Nov 2026*

STRENGTHS

- Systems architecture
- Product thinking
- Solo end-to-end delivery
- Technical leadership
- Debugging under production pressure
- Documentation
- Systems design for any scale — offline to high-load
- Adaptive architecture: low-cost to enterprise-scale
- Building for constraints and building for scale